SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

Code of Conduct

I C.pavani/(192524260) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: c.pavani

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

20/07/26

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CO_1 - ASSESSMENT TOOL 1

17/25

SHORT ANSWER TEXT

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Course name : cloud computing
Department : CSE [AI &DS]
Course code : CSA1502

1) cloud computing and explain the any four characteristics and that make it different from of their tradional on-premise computing.

Definition :

Cloud computing is the delivery of computing services such as servers, storage databases, networking software, and anylatics. over the internets. ("The cloud") on pay as you use basis instead of owning. and maintaing physical infrastructures.

Characteristics :

on-demand network self access :-

- users can access computing resources whenever required without contacting the service which provider. can provides their network in to the on-demand network of self access.

Example :-

Creating a virtual machine on AWS within minutes.

Board network access :

- Cloud services are available over the internet and can be accessed from their computers, tablets, and their smartphones.
- This enables remote work and global accessibility.

Research poling :

- Cloud provides shape physical resources among multiple customers using virtulization.
- Resources are dynamically assigned accordingly to customer demand.

Rapid electricity :

- Resources can be increased or decreased automatically depending on workload

Example :- An e-comemerce website which increases server capacity during festive sales and reduces it afterward.

2. The evolution of their cloud computing from hardware evolution to internet software evolution.

Cloud Computing evolved through several technological advancements.

1. <u>Hardware Evolution</u>

- Early computers were large and expensive mainframes.
- Later, personal computers and powerful serves became available.

2. <u>Virtualization</u>.

- Virtualization allowed multiple virtual machines to run on a single physical server.
- This improved hardware utilization and reduced costs.
- It became the foundation of cloud computing.

3. <u>Internet Evolution</u>.

- Growth of broadband internet enabled reliable high speed communication.
- users could access remote servers from anywhere.

4. Software Evolution

- Applications shifted from desktop software to web based software.
- Technologies such as Software as a Service (SaS) became popular.

5. Modern Cloud platforms.

Companies such as Amazon web services, microsoft Azure, and google cloud provide scalable cloud services.

Users can rent infrastructure, platforms, and software over the internet.

Automation and virtualization make cloud services efficient and cost-effective.

Conclusion :

The evolution of hardware virtualization, internet connectivity and web based software collectively enabled modern cloud computing platforms.

The role of server virtualization in their cloud computing. and then differentiate the between of virtualization as an enabling technology and their cloud computing a. Service model.

* The server virtualization in their cloud computing is the physical and their virtualization. of their technologies in the computing of their cloud.
* The cloud computing of their server virtualization is which allows the single physical server to be the which divides multiple virtualization server. (MVM).
* where it was edge of their computing of cloud each one operators like as single machine in their cloud computing.

Rules :
Biffer browser utilization :-
The physical server can host mulutiple servers.
Ex :- cpu memory storage.

(ii) cost reduction :

The reduction is happen in which multiple servers to lowering their memory of storage.

3

4) The role of data centers in delivering their cloud services..

- A data center is facility that contains Servers storage systems networking equipment power supply cooling systems and security infrastructure used to deliver cloud services.

Rules of data centres :-

1. Storage of data :

- Stores customer files databases, applications and backups.
- It ensures the secure and reliable data structures.

2. Hosting application :

- Rules Runs website, business applications and cloud computing
- Provides continuous avaliability.

3. computing power :

- It provides processing resources for cloud applications.
- The Supports artifical intkgence machine learning and big data processing.

4. High availability :

- Minimizes down time

5. Security :

- physical Security through. Surveliance and restricted access.

6. Disaster recovery :

- The ensures of these business continuously during failures.

5) Differentiate loas, paas, xaas one.

model	meaning	model	Real world Example.
lass	Infastructure as service system.		Amazon web Services
pass			
xass.			
sass.			